

Translative covering densities of order $n \log n$

OpenAI

September 23, 2026

Abstract

For every sufficiently large dimension n , we construct a centrally symmetric convex body whose translative covering density exceeds $cn \log n$, where $c > 0$ is absolute. This disproves the existence of a universal linear upper bound and matches the order of Rogers' upper bound.

Contents

1	Introduction	2
1.1	Antecedents and the additional uniformity	3
1.2	Proof outline	3
1.3	Notation	4
2	The random body	4
3	Reducing all periodic coverings to finitely many patterns	5
4	Intersections of spherical caps	9
5	Grouping cap conditions	13
6	Poisson witnesses and their lower tail	16
6.1	The finite inequality	16
6.2	Measurable slots and matching diagrams	17
6.3	Self and cross diagrams	19
6.4	Application to residual caps	20
7	Many targets for each finite pattern	22
7.1	A deterministic choice of targets	22
7.2	From targets to a small covering probability	23
8	Completion of the proof	25
8.1	The optimal order for translative and lattice coverings	26
9	Rational polytopes and normal lists	26
A	Covering density and periodization	28

1 Introduction

A convex body in $\mathbb{R}^n$ is a compact convex set with nonempty interior. A set $X \subset \mathbb{R}^n$ is *locally finite* if every bounded set contains only finitely many of its points. It gives a *translative covering* of K when $K + X = \mathbb{R}^n$. Write $Q_R = [-R, R]^n$. We define the translative covering density by

$$\theta_T(K) = \inf_{\substack{X \text{ locally finite} \\ K+X=\mathbb{R}^n}} \text{vol}(K) \limsup_{R \rightarrow \infty} \frac{\#(X \cap Q_R)}{(2R)^n}. \quad (1.1)$$

Thus the density measures the asymptotic volume of translates per unit volume, counting overlaps. It is not the fraction of space covered, which is one for every covering. No asymptotic density limit is assumed for an individual arrangement.

A *periodic translative covering* is a covering

$$\mathbb{R}^n = K + \Lambda + T,$$

where Λ is a full-rank lattice and T is a finite set of distinct representatives modulo Λ . Its center intensity is $|T|/\det \Lambda$, and its density is $|T| \text{vol}(K)/\det \Lambda$. Put

$$\theta_{\text{per}}(K) = \inf \left\{ \frac{|T| \text{vol}(K)}{\det \Lambda} : K + \Lambda + T = \mathbb{R}^n \right\}.$$

The infimum ranges over every lattice and every finite set T . In particular, there is no bound on the number of cosets. Allowing repeated representatives of the same coset gives the same infimum: deleting redundant representatives preserves the covering and does not increase its listed density. When we later retain indexed center lists, we count such repetitions explicitly. Both infima are finite: an interior cube of K gives a covering by translates on a sufficiently fine cubic lattice.

The classical periodization principle gives

$$\theta_T(K) = \theta_{\text{per}}(K). \quad (1.2)$$

We prove the precise statement in Appendix A, including agreement with lower or upper centered-ball density and average covering multiplicity. See also Prosanov [11, Section 1.1 and Theorem 2.1], who attributes the principle to Rogers. These are equalities of optimal densities; the lower and upper densities of an individual nonperiodic arrangement need not agree.

Theorem 1.1. *There are absolute constants $c > 0$ and n_0 such that, for every integer $n \geq n_0$, some centrally symmetric convex body $K_n \subset \mathbb{R}^n$ satisfies*

$$\theta_T(K_n) > cn \log n.$$

Thus there is no absolute constant C such that $\theta_T(K) \leq Cn$ for every convex body in every dimension, even if the bodies are required to be centrally symmetric. This answers negatively the universal linear-bound question recorded by Naszódi [9, Section 3]. Rogers [12] proved that every convex body in dimension $n \geq 3$ has a periodic translative covering of density at most

$$n \log n + n \log \log n + 5n. \quad (1.3)$$

Together with Theorem 1.1, this shows that the supremum of covering density over convex bodies has order $n \log n$. The same order holds when one restricts the bodies to be centrally symmetric. The constant in Theorem 1.1 is not optimized. The bodies may moreover be chosen to be centrally symmetric polytopes with rational vertices, with the same constant c ; see Corollary 9.1.

1.1 Antecedents and the additional uniformity

Rogers' theorem supplies a universal upper bound for arbitrary convex bodies. Fejes Tóth [4] sharpened its lower-order term to obtain density $n(\log n + \log \log n + 1 + o(1))$ using only $O(\log n)$ cosets of a common lattice. A finite union of lattice cosets has substantially more freedom than one lattice, even with this bound on the number of cosets. On the lower-bound side, the classical work of Coxeter, Few and Rogers [2] gives an order- n obstruction already for unrestricted coverings by equal Euclidean balls. For Euclidean balls, Dumer [3, Corollary 2] gives the upper bound $(1/2 + o(1))n \log n$. The gap for Euclidean balls remains; our theorem determines the optimal order when the convex body may also be chosen.

The proof builds on the random-slab construction and Poisson witness method of Li and Liu [8, Theorem 1.3]. They prove an order- $n \log n$ lower bound for *lattice* covering density, in which the centers form a single lattice. Their common-intersection estimates, terminal clustering of caps, and overlap-diagram bounds provide the local probabilistic machinery used here. We give the necessary versions and their proofs, including the continuous form of Janson's inequality.

The additional issue is to exclude every finite union of lattice cosets. A net of lattice bases cannot encode an arbitrary number of cosets. We instead localize an arrangement in a fixed Euclidean window, keeping every center with its multiplicity, and round the resulting list to a finite grid. Periodic averaging controls this list even when the lattice has very short periods or a highly skew fundamental domain. The resulting finite class is fixed before the body is sampled, so it includes arrangements chosen after the body is known. A sufficiently strong lower-tail bound then excludes all of them simultaneously.

For comparison, Bukh, Gao, Liu, Pikhurko, and Sun [1, Theorem 1.3] show that, in their packing-torus model, Poisson centers at a specified density of order $n \log n$ fail to cover a given body with probability tending to one. Their random-centers barrier has different quantifiers: an exceptional arrangement might still cover efficiently. Here the body is random and the conclusion excludes every periodic arrangement for one realization of that body.

1.2 Proof outline

Intersect a slightly enlarged Euclidean ball with symmetric slabs whose unit normals are sampled by a Poisson process. The normalization gives this body a substantial volume with positive probability. A hypothetical covering of low center intensity supplies, after localization and rounding, a finite center list with many suitable target points in a small window. We must show that each such list is very unlikely to cover that window.

At a target y , only centers p within the outer ball's radius of y can cover it. For each such center, the residual vector $y - p$ determines the antipodal caps of slab normals that would remove y from that translate. A hole appears if the Poisson process hits at least one cap in each pair. Nearly coincident cutting sets are strongly dependent. Following Li and Liu, we group these conditions into blocks and choose one common subset of normals for each block. Assigning distinct Poisson points to these subsets certifies a hole. The block construction controls the complete overlap sum of these certificates, including their self-overlaps.

The targets are chosen at two angular scales. For every pair of distinct targets, every pair of their relevant residual directions is separated at a scale of order $\sqrt{\log n/n}$, identifying opposite directions because the slabs are symmetric. The cap estimates then bound the dependence between their hole certificates. The body lies in a ball whose radius is only slightly larger than the slab half-width, so its cutting caps have a small fixed angular radius. At a larger fixed separation, all cutting caps belonging to one target are disjoint from those belonging to the other. A cylinder-volume estimate

lets us arrange this stronger separation for all but an exponentially small fraction of target pairs, while retaining the finer separation for every pair.

Let R denote the density lower bound sought. We can choose $\exp(c_1 n \log n)$ targets for a fixed $c_1 > 0$, enough to absorb a self-overlap cost $\exp(CR)$ when $R = cn \log n$ and c is sufficiently small. The two angular scales control the remaining overlap between distinct targets. Janson's lower-tail inequality turns these bounds into a covering probability that is doubly exponentially small in n , with decay strong enough to dominate the number of rounded patterns. A second-moment bound would not give the required decay.

Section 2 defines the body and the constants. Section 3 constructs the finite pattern class. Sections 4–6 establish the cap, block, and Poisson estimates. Section 7 proves the fixed-pattern bound, and Section 8 completes the proof. Section 9 gives rational-polytope and rational-normal versions of the construction. Appendix A proves the density equivalence that transfers the periodic exclusion to unrestricted coverings.

1.3 Notation

Logarithms are natural, and $[t]_+ = \max\{t, 0\}$. We write $|x|$ for Euclidean norm, $B_r = \{x \in \mathbb{R}^n : |x| \leq r\}$, $B = B_1$, and $\omega_n = \text{vol}(B)$. For measurable sets in $\mathbb{R}^n$, $|A|$ also denotes Lebesgue volume; for finite lists and index sets it denotes cardinality. Surface measure σ on S^{n-1} is normalized to have total mass one. For nonzero vectors we use the projective angle

$$\angle(x, z) = \arccos \frac{|x \cdot z|}{|x||z|} \in [0, \pi/2].$$

Lists are indexed: two entries can occupy the same point or define the same cap. Every sum and cardinality over such a list retains its multiplicities. Constants implicit in $O(\cdot)$ depend only on parameters explicitly fixed before n tends to infinity.

2 The random body

Fix the numerical constants

$$\varepsilon = 2^{-30}, \quad a = 1 + \varepsilon, \quad D = 2^{-10} = \varepsilon^{1/3}, \quad v = \frac{1}{64}. \quad (2.1)$$

For each sufficiently large n , put

$$\eta = n^{-4}, \quad b = a(1 + 1/n), \quad r_- = a(1 - 10 \log n/n). \quad (2.2)$$

The small window is B_D . Its size relative to a will govern the number of patterns, through

$$\lambda_0 = \log(1 + D/a), \quad \gamma = \log \frac{D}{8\sqrt{\varepsilon}} - 2\lambda_0 = \log 4 - 2\lambda_0. \quad (2.3)$$

Since $\lambda_0 < D = 2^{-10}$, we have $\gamma > 3\lambda_0$. These constants are fixed; none tends to zero with dimension.

Set

$$M = (\sigma\{u : |u \cdot e_1| > 1/a\})^{-1}, \quad \mu = M\sigma, \quad (2.4)$$

and let Π be a Poisson process on S^{n-1} with intensity μ . Equivalently, sample a Poisson random integer of mean M , then that many independent σ -distributed unit normals. Define

$$K = bB \cap \bigcap_{u \in \Pi} \{x : |u \cdot x| \leq 1\}, \quad (2.5)$$

$$K^+ = (b + 3\eta)B \cap \bigcap_{u \in \Pi} \{x : |u \cdot x| \leq 1 + 3\eta\}. \quad (2.6)$$

The enlarged body K^+ absorbs the displacement of centers during rounding. Since $M < \infty$, the list of normals is finite almost surely. Both bodies are compact and centrally symmetric, and $B \subset K$, so K has nonempty interior.

Lemma 2.1. *The body in (2.5) satisfies*

$$\mathbb{P}\{|K| > 2v\omega_n a^n\} \geq \frac{\exp(-1) - 2v}{3 - 2v} > 0.$$

Proof. If $|x| \leq a$, the μ -measure of normals with $|u \cdot x| > 1$ is at most one by rotational invariance and (2.4). The probability that there is no such Poisson normal is therefore at least $\exp(-1)$. Integrating over aB gives $\mathbb{E}|K| \geq \exp(-1)\omega_n a^n$. On the other hand, $|K| \leq \omega_n b^n < 3\omega_n a^n$. Thus $X = |K|/(\omega_n a^n)$ has $0 \leq X \leq 3$ and $\mathbb{E}X \geq \exp(-1)$. Writing $p = \mathbb{P}(X > 2v)$, we have $\mathbb{E}X \leq 2v(1 - p) + 3p$, which proves the claim. $\square$

We shall rule out coverings with center intensity at most

$$\rho_0 = \frac{R}{2v\omega_n a^n}, \quad (2.7)$$

where R is a positive parameter. This threshold does not depend on the realized volume. Once every such covering is excluded, the definition of θ_{per} gives $\theta_{\text{per}}(K) \geq \rho_0|K|$. The strict volume event in Lemma 2.1 will then imply $\theta_{\text{per}}(K) > R$, including strictness at the infimum.

3 Reducing all periodic coverings to finitely many patterns

We first remove the lattice and its fundamental domain from the subsequent probability estimate. A periodic arrangement of bounded center intensity has a translated ball containing relatively few centers and many points with controlled residual vectors. Rounding those centers produces a list in a fixed finite family. The argument counts every occurrence of a center, so it applies to arbitrarily short or skew periods and to coinciding cosets.

Use the fixed parameters a, D, v, b, r_-, η from the preceding section, and let $1 \leq R \leq n^2$. Define

$$C_{\text{loc}} = \frac{1000}{v}, \quad w_a(r) = 1 + n[1 - r^2/a^2]_+ \quad (r \geq 0), \quad h_n = \frac{\eta}{100n}.$$

The weight w_a records the radial cost of a center near a target point. Our finite alphabet and maximum list length are

$$\mathcal{Q}_n = h_n \mathbb{Z}^n \cap B_{D+b+5\eta}, \quad N_n = \left\lceil \frac{20R}{v} \left(\frac{D+b+5\eta}{a} \right)^n \right\rceil. \quad (3.1)$$

An ordered list $P = (p_1, \dots, p_m)$ over $\mathcal{Q}_n$ may contain repeated entries. Define $G_P \subseteq B_D$ to consist of the points y satisfying the following three conditions:

$$|y - p_j| \geq r_- - 2\eta \quad (1 \leq j \leq m), \quad (3.2)$$

$$\#\{j : |y - p_j| \leq b + 3\eta\} \leq 2C_{\text{loc}}R, \quad (3.3)$$

$$\sum_{j: |y - p_j| \leq b + 3\eta} w_a(|y - p_j|) \leq 2C_{\text{loc}}R. \quad (3.4)$$

The count condition follows from the weight condition because $w_a \geq 1$; we display both to identify the cardinality and cost bounds used later. All sums count indices, including repeated entries. Set

$$\mathcal{F}_n = \left\{ P \in \bigcup_{m=0}^{N_n} \mathcal{Q}_n^m : |G_P| \geq \frac{1}{4}|B_D| \right\}. \quad (3.5)$$

The dependence of N_n and $\mathcal{F}_n$ on R is suppressed in the notation. Empty lists are allowed, with all their sums equal to zero.

Proposition 3.1 (Deterministic localization). *For all sufficiently large n , uniformly for $1 \leq R \leq n^2$, the family $\mathcal{F}_n$ has the following properties. With $\lambda_0 = \log(1 + D/a)$,*

$$N_n \leq n^{C_1} \exp(\lambda_0 n), \quad \log |\mathcal{F}_n| \leq \exp(\lambda_0 n + O(\log n)), \quad (3.6)$$

where C_1 and the implicit constant depend only on the fixed parameters.

Let $\mathcal{U} \subset S^{n-1}$ be any finite set, and put

$$K = bB \cap \bigcap_{u \in \mathcal{U}} \{x : |u \cdot x| \leq 1\}, \quad K^+ = (b + 3\eta)B \cap \bigcap_{u \in \mathcal{U}} \{x : |u \cdot x| \leq 1 + 3\eta\}.$$

Suppose that Λ is any full-rank lattice, $T \subset \mathbb{R}^n$ is finite, and

$$\frac{|T|}{\det \Lambda} \leq \rho_0 := \frac{R}{2v\omega_n a^n}.$$

If $K + \Lambda + T$ covers $\mathbb{R}^n$, there is a list $P = (p_1, \dots, p_m) \in \mathcal{F}_n$ such that

$$B_D \subseteq \bigcup_{j=1}^m (p_j + K^+). \quad (3.7)$$

The constructed list in fact satisfies $|G_P| \geq |B_D|/2$.

For every $P \in \mathcal{F}_n$ and $y \in G_P$, list all residual vectors $x_i = y - p_j$ with $|y - p_j| \leq b + 3\eta$, retaining their indices, and denote their number by m_y . Then

$$\begin{aligned} r_- - 2\eta \leq |x_i| \leq b + 3\eta, \quad m_y \leq 2C_{\text{loc}}R, \quad \sum_{i=1}^{m_y} w_a(|x_i|) \leq 2C_{\text{loc}}R, \\ m_y + \sum_{i=1}^{m_y} w_a(|x_i|) \leq 4C_{\text{loc}}R. \end{aligned} \quad (3.8)$$

Proof. We prove the localization and rounding assertions first, then count the resulting lists. All dimension thresholds in the proof are uniform for $1 \leq R \leq n^2$.

Averaging the full multiset of centers. Give each center an index $(t, \ell) \in T \times \Lambda$, at position $t + \ell$. Distinct indices remain distinct even if they have the same position. This multiset is locally finite because T is finite and Λ is discrete. Write $\rho = |T|/\det \Lambda$, and let F be a fundamental parallelepiped of Λ . For every nonnegative integrable function f ,

$$\frac{1}{|F|} \int_F \sum_{(t, \ell) \in T \times \Lambda} f(y - t - \ell) dy = \rho \int_{\mathbb{R}^n} f(x) dx. \quad (3.9)$$

Indeed, Tonelli's theorem permits the sums to be interchanged with the integral. For each t , the sets $F - t - \ell$, $\ell \in \Lambda$, partition $\mathbb{R}^n$ up to null sets. This proves the identity without any restriction on the shape or size of F .

Put $\beta = b + 4\eta$. Call a point $y \in \mathbb{R}^n$ unrounded-good if there is no center occurrence at distance less than r_- , at most $C_{\text{loc}}R$ occurrences have distance at most β , and

$$\sum_{(t, \ell): |y - t - \ell| \leq \beta} w_a(|y - t - \ell|) \leq C_{\text{loc}}R. \quad (3.10)$$

Let G be this periodic measurable set. These conditions respectively exclude short residuals, bound the number of relevant residuals, and bound their total radial cost. Using the full ball ensures that every index entering a rounded test is counted before rounding.

For a uniform point modulo Λ , the expected number of occurrences in its open r_- -ball is at most

$$\rho \omega_n r_-^n \leq \frac{R}{2v} \left(1 - \frac{10 \log n}{n}\right)^n \leq \frac{Rn^{-10}}{2v} \leq \frac{1}{200}$$

for large n . Hence the first condition fails with probability at most $1/200$. Also

$$\left(\frac{\beta}{a}\right)^n \leq \exp(1 + 4/n^3) < 3$$

for large n , so the expected outer count is at most $3R/(2v)$. For the weighted condition, polar integration gives the exact identity

$$\begin{aligned} \int_{B_\beta} w_a(|x|) dx &= \omega_n \beta^n + n \int_{B_a} (1 - |x|^2/a^2) dx \\ &= \omega_n \beta^n + \frac{2n}{n+2} \omega_n a^n < 5\omega_n a^n. \end{aligned} \quad (3.11)$$

Equation (3.9) therefore bounds the expected weighted sum by $5R/(2v)$. Markov's inequality bounds the two remaining failure probabilities by $3/2000$ and $5/2000$, respectively. Consequently

$$\frac{|G \cap F|}{|F|} \geq 1 - \frac{1}{200} - \frac{3}{2000} - \frac{5}{2000} > \frac{99}{100}. \quad (3.12)$$

One window with both required bounds. For z uniform modulo Λ , let $M(z)$ count all center occurrences in $z + B_{D+b+4\eta}$. Equations (3.9) and (3.1) imply

$$\mathbb{E} M(z) \leq \frac{R}{2v} \left(\frac{D+b+4\eta}{a}\right)^n \leq \frac{N_n}{40}.$$

Moreover, periodicity and Fubini's theorem give

$$\mathbb{E} \frac{|(z + B_D) \setminus G|}{|B_D|} = 1 - \frac{|G \cap F|}{|F|} \leq \frac{1}{100}.$$

Both expressions concern the entire Euclidean window, including all its overlapping period copies. Markov's inequality gives

$$\mathbb{P}\{M(z) > N_n\} \leq \frac{1}{40}, \quad \mathbb{P}\{|(z + B_D) \setminus G| > |B_D|/2\} \leq \frac{1}{50}.$$

The sum is less than one. Fix a slide z outside these two exceptional events, and retain every occurrence in $z + B_{D+b+4\eta}$. There are at most N_n such occurrences, and at least half of $z + B_D$ lies in G .

Rounding and transferring coverage. Translate this window and the retained centers by $-z$, and round every center coordinatewise to the nearest point of $h_n\mathbb{Z}^n$, using a fixed rule for ties. Each center moves a distance at most

$$\delta_n = \frac{\sqrt{n}h_n}{2} = \frac{\eta}{200\sqrt{n}} \leq \frac{\eta}{200}. \quad (3.13)$$

The rounded centers belong to $\mathcal{Q}_n$; enumerate them as an ordered list P , retaining all repeated entries.

We check that every $y \in B_D$ with $y + z \in G$ belongs to G_P . Every unrounded residual has length at least r_- , so every rounded one has length at least $r_- - \delta_n \geq r_- - 2\eta$. A rounded residual of length at most $b + 3\eta$ comes from an unrounded residual of length at most

$$b + 3\eta + \delta_n < b + 4\eta.$$

Thus at most $C_{\text{loc}}R$ rounded indices are relevant. This strict separation of the two outer radii ensures that every index entering a rounded test was counted before rounding.

The function w_a is globally Lipschitz on $[0, \infty)$ with constant $2n/a$. Its derivative has absolute value $2nr/a^2 \leq 2n/a$ for $0 < r < a$, and it is constant for $r > a$. If r and r' are the unrounded and rounded lengths of a relevant residual, then

$$w_a(r') \leq w_a(r) + \frac{2n\delta_n}{a} \leq \left(1 + \frac{2n\delta_n}{a}\right) w_a(r),$$

where the last step uses $w_a(r) \geq 1$. Summing over the relevant indices, all of which were included in (3.10), gives

$$\sum_{j: |y - p_j| \leq b+3\eta} w_a(|y - p_j|) \leq \left(1 + \frac{2n\delta_n}{a}\right) C_{\text{loc}}R \leq 2C_{\text{loc}}R.$$

This verifies (3.2)–(3.4). In particular, $|G_P| \geq |B_D|/2$ and $P \in \mathcal{F}_n$.

Now assume that the original periodic arrangement covers by K . For any point of $z + B_D$, every center whose translate of K contains that point lies in $z + B_{D+b}$ because $K \subseteq bB$. Such a center was retained. Moving it by at most δ_n changes the residual norm by at most δ_n and each scalar product with a unit normal by at most δ_n . Hence its rounded translate of K^+ contains the translated point of B_D . This proves (3.7) for every point of the closed window.

Counting the lists. Write $J_n = |\mathcal{Q}_n|$. A containing cube gives

$$J_n \leq \left(1 + \frac{2(D + b + 5\eta)}{h_n}\right)^n, \quad \log J_n = O(n \log n).$$

Since $b = a(1 + 1/n)$, $\eta = n^{-4}$, and $1 \leq R \leq n^2$,

$$\log N_n = \lambda_0 n + O(\log n).$$

This proves the first bound in (3.6). Counting ordered lists allows all multiplicities and gives

$$|\mathcal{F}_n| \leq \sum_{m=0}^{N_n} J_n^m \leq (N_n + 1) J_n^{N_n}.$$

The two preceding estimates give the second bound in (3.6). Finally, (3.8) follows directly from the definition of G_P and $w_a \geq 1$. $\square$

The family $\mathcal{F}_n$ depends only on the fixed numerical parameters, n , and R . Proposition 3.1 holds for every choice of the slab normals and every periodic covering of the required intensity. Thus a probability bound uniform over this finite family also controls centers chosen after the body is known. The next sections obtain such a bound by finding a point of G_P missed by all translates in a fixed list.

4 Intersections of spherical caps

We need two properties of the cutting caps: separated caps have small overlap, whereas a collection of nearby caps has a substantial common intersection. For a target $y \in G_P$ and a relevant center p , write $x = y - p$. The set

$$C(x) = \{u \in S^{n-1} : |u \cdot x| > 1 + 3\eta\}$$

consists of the normals whose slabs exclude y from $p + K^+$. Figure 1 illustrates this correspondence. The estimates below allow the slightly more general thresholds needed to control these caps uniformly.

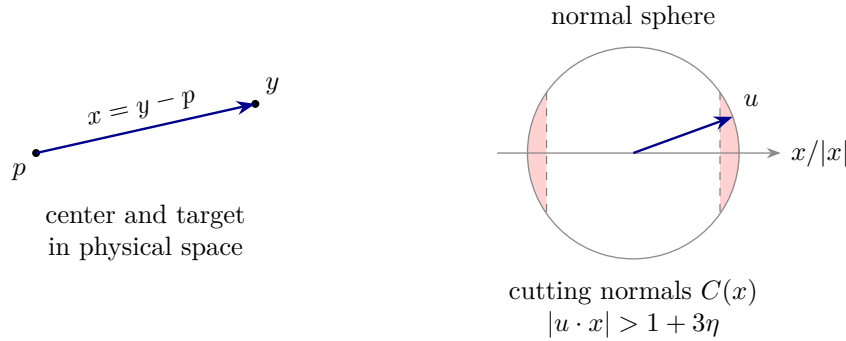


Figure 1: Plane-section schematic of a residual and its cutting cap. The shaded antipodal caps lie beyond the two threshold hyperplanes $u \cdot x = \pm(1 + 3\eta)$. A sampled normal in $C(x)$ excludes the target y from $p + K^+$. The drawing does not depict high-dimensional cap measure. For a block of residuals, one normal in their common subset cuts them all; assigning distinct normals to all blocks is a sufficient certificate of a hole.

The estimates adapt Li-Liu's cap and common-lens arguments [8, Lemma 6.1 and Lemmas 9.2–9.3]. In our thin radial shell, a bisector argument gives a short proof of the overlap estimate with a constant prefactor. Keeping that prefactor independent of dimension will be important when we sum overlaps.

Lemma 4.1 (Cap estimates). *Fix $1 < a < 2$ and $A, Q, B_0 > 0$. There exist constants $C_a, c'_a > 0$, $C_{a,A} > 0$, and $L = L(a) > 0$ such that the following holds for all sufficiently large n , depending on a, A, Q, B_0 . Let σ be normalized surface measure on S^{n-1} , and set*

$$M = \sigma\{u : |u \cdot e_1| > 1/a\}^{-1}, \quad \mu = M\sigma.$$

Let $x_1, \dots, x_m \in \mathbb{R}^n$ and $\epsilon_1, \dots, \epsilon_m \in \mathbb{R}$, where $1 \leq m \leq n^2$, satisfy

$$a \left(1 - \frac{A \log n}{n}\right) - Qn^{-4} \leq |x_i| \leq a(1 + 1/n) + Qn^{-4}, \quad |\epsilon_i| \leq Qn^{-4}.$$

Define

$$C_i = \{u \in S^{n-1} : |u \cdot x_i| > 1 + \epsilon_i\}, \quad c_i = \mu(C_i), \quad \bar{c}_i = \min(c_i, 1/2), \quad s_i = -\log \bar{c}_i,$$

and, using projective angles, define

$$d(i, j) = \sqrt{n} \angle(x_i, x_j) + \sqrt{|s_i - s_j|}, \quad r(S) = \min_{k \in S} \max_{i \in S} d(k, i)^2$$

for each nonempty set S of indices. Then

$$n^{-C_{a,A}} \leq c_i \leq C_a, \quad s_i \leq C_a(1 + n[1 - |x_i|^2/a^2]_+), \quad (4.1)$$

$$\frac{\mu(C_i \cap C_j)}{\sqrt{\bar{c}_i \bar{c}_j}} \leq C_a \exp\{-c'_a d(i, j)^2\}. \quad (4.2)$$

The function d is a pseudometric on the indexed list. If $r(S) \leq B_0 \log n$ and $i_0 \in S$ minimizes the raw intensity c_i , then

$$\mu\left(\bigcap_{i \in S} C_i\right) \geq e^{-L(1+r(S)+\log |S|)\bar{c}_{i_0}}. \quad (4.3)$$

In particular, L is independent of A, Q, B_0 ; these parameters may affect the starting dimension.

Proof. We first record one-dimensional estimates, then prove the overlap and common-intersection bounds. Constants with a subscript a below may change from line to line, but depend only on a .

One-cap probabilities and radial costs. Write $t_* = 1/a$, $e_i = x_i/|x_i|$, and $\tau_i = (1 + \epsilon_i)/|x_i|$. For a unit vector e , let

$$F_n(t) = \sigma\{u : u \cdot e > t\} = k_n \int_t^1 (1 - v^2)^{(n-3)/2} dv, \quad 0 < t < 1,$$

where k_n is the normalization constant. Substitution of $z = 1 - v^2$, using $1 \leq 1/v \leq 1/t$, gives

$$\frac{k_n}{n-1} (1 - t^2)^{(n-1)/2} \leq F_n(t) \leq \frac{k_n}{(n-1)t} (1 - t^2)^{(n-1)/2}. \quad (4.4)$$

Consequently,

$$\frac{(n-1)t}{1-t^2} \leq -\frac{d}{dt} \log F_n(t) \leq \frac{n-1}{1-t^2}. \quad (4.5)$$

These exact inequalities avoid dimension-dependent prefactors in ratios of cap probabilities.

All the thresholds satisfy

$$\tau_i = t_* + O_{a,A,Q}(\log n/n), \quad \tau_i \geq t_* - C_a/n, \quad (4.6)$$

for sufficiently large n , and hence belong to the fixed interval $J = [t_*/2, (1 + t_*)/2] \subset (0, 1)$. Since $M = (2F_n(t_*))^{-1}$, we have $c_i = F_n(\tau_i)/F_n(t_*)$. The logarithmic derivative in (4.5) is bounded by $C_a n$ on J . This proves $c_i \leq C_a$, while the lower radial bound proves $c_i \geq n^{-C_{a,A}}$. More explicitly, putting $\delta_i = [1 - |x_i|^2/a^2]_+$, elementary algebra gives

$$\begin{aligned} (\tau_i - t_*)_+ &\leq C_a(\delta_i + Qn^{-4}), & (t_* - \tau_i)_+ &\leq C_a(n^{-1} + Qn^{-4}), \\ n\delta_i &\leq C_{a,A} \log n + O_{a,Q}(n^{-3}). \end{aligned}$$

Integrating (4.5) therefore gives $-\log c_i \leq C_a(1 + n\delta_i)$, after increasing the starting dimension so that $Qn^{-3} \leq 1$. As $s_i = \max(\log 2, -\log c_i)$, this proves (4.1). The additive constant is needed even when $c_i \geq 1/2$.

The overlap bound. Consider two ordinary half caps with thresholds τ, v drawn from our list and with axis angle $\theta \in [0, \pi]$. Put $t = (\tau + v)/2$. If $\theta < \pi$, adding their two defining inequalities shows that their intersection lies in the half cap along the angle bisector with threshold

$$q = t / \cos(\theta/2).$$

The intersection is empty if $\theta = \pi$ or $q \geq 1$, so suppose $q < 1$. Let $f(z) = \log(1 - z^2)$. Its second derivative is bounded in absolute value on J , whence

$$f(t) - \frac{1}{2}f(\tau) - \frac{1}{2}f(v) \leq C_a(\tau - v)^2.$$

Furthermore,

$$\frac{1 - q^2}{1 - t^2} = 1 - \frac{t^2}{1 - t^2} \tan^2(\theta/2) \leq e^{-c'_a \theta^2},$$

using $t \geq t_*/2$ and $\tan(\theta/2) \geq \theta/2$. Apply the upper bound in (4.4) to $F_n(q)$ and the lower bound to both terms in the following denominator:

$$\frac{F_n(q)}{\sqrt{F_n(\tau)F_n(v)}} \leq q^{-1} \exp\{C_a n(\tau - v)^2 - c'_a n \theta^2\} \leq C_a e^{-c'_a n \theta^2}. \quad (4.7)$$

Indeed, $q^{-1} \leq 2/t_*$, and $n(\tau - v)^2 = O_{a,A,Q}((\log n)^2/n) \rightarrow 0$ by (4.6). This last fact is the benefit of the thin shell. The argument does not require $q \in J$, so it also applies when the bisector threshold approaches 1.

For projective angle $\phi = \angle(x_i, x_j)$, the four pairs of signed half caps have ordinary angles ϕ or $\pi - \phi \geq \phi$. Summing (4.7) over these four disjoint sign choices and using $c_i = 2MF_n(\tau_i)$ gives

$$\mu(C_i \cap C_j) \leq C_a \sqrt{c_i c_j} e^{-c'_a n \phi^2}.$$

By (4.1), $c_i/\bar{c}_i \leq \max(1, 2C_a)$, so clipping the denominator costs only a constant. Independently, the trivial intersection bound gives

$$\frac{\mu(C_i \cap C_j)}{\sqrt{\bar{c}_i \bar{c}_j}} \leq C_a e^{-|s_i - s_j|/2}.$$

To see this, order the clipped intensities as $\bar{c}_i \leq \bar{c}_j$. If $\bar{c}_i < 1/2$, use $\mu(C_i \cap C_j) \leq c_i = \bar{c}_i$; otherwise both clipped intensities equal $1/2$, and use $c_i \leq C_a$. The minimum of the two bounds is at most their geometric mean. Together with

$$d(i, j)^2 \leq 2(n\phi^2 + |s_i - s_j|),$$

this proves (4.2).

Projective angle satisfies the triangle inequality, as the quotient of spherical distance under antipodal identification. The inequality $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ shows that $\sqrt{|s_i - s_j|}$ also satisfies it. Thus d is a pseudometric; distinct labels, including repeated caps, may have distance zero.

A common intersection. Set $r = r(S)$ and $m_S = |S|$, and choose an index attaining the minimum defining r . By the triangle inequality, each axis in S has projective angle at most $2\sqrt{r/n}$ from $e_0 = e_{i_0}$. Since F_n is strictly decreasing, the raw-minimum anchor has the largest threshold: $\tau_0 = \tau_{i_0} \geq \tau_i$. Orient each axis so that its ordinary angle ϕ_i with e_0 equals its projective angle, and write

$$e_i = (\cos \phi_i)e_0 + (\sin \phi_i)v_i, \quad \Phi = \max_{i \in S} \phi_i \leq 2\sqrt{r/n},$$

where v_i is a unit tangent vector. For $\phi_i = 0$, its choice is immaterial.

We use the elementary bound

$$\mathbb{P}(W \cdot v < -h) \leq e^{-dh^2/2} \quad (4.8)$$

for a uniform $W \in S^{d-1}$, $d \geq 2$, any unit v , and $h \geq 0$. Here is a proof. Rotational invariance reduces to $X = W_1$. Integrating the derivative of $x^{2j-1}(1-x^2)^{(d-1)/2}$ on $[-1, 1]$ shows that

$$\mathbb{E}X^{2j} = \frac{2j-1}{d+2j-2} \mathbb{E}X^{2j-2} \leq \frac{(2j)!}{2^j j! d^j}.$$

Odd moments vanish. The exponential series therefore gives $\mathbb{E}e^{zX} \leq e^{z^2/(2d)}$; applying Markov's inequality with $z = -dh$ proves (4.8).

Conditionally on $u \cdot e_0 = t$, a uniform sphere point has the form $u = te_0 + \sqrt{1-t^2}W$, with W uniform on the tangent sphere S^{n-2} . Put

$$h = \sqrt{\frac{2 \log(2m_S)}{n-1}}, \quad \delta = 2\Phi^2 + 4\Phi h.$$

By (4.8) and a union bound, with conditional probability at least $1/2$ one has $W \cdot v_i \geq -h$ for every $i \in S$. For large enough n , $\cos \Phi \geq 1/2$ and $\tau_0 + \delta \in J$, because $r \leq B_0 \log n$ and $m_S \leq n^2$. On that tangent event, whenever $t > \tau_0 + \delta$,

$$u \cdot e_i \geq t \cos \phi_i - h \sin \phi_i > \tau_0 - \Phi^2/2 + \delta/2 - h\Phi \geq \tau_0 \geq \tau_i.$$

Thus all these points lie in every double cap. Integration over the axial coordinate gives

$$\mu \left(\bigcap_{i \in S} C_i \right) \geq \frac{M}{2} F_n(\tau_0 + \delta) = \frac{c_{i_0}}{4} \frac{F_n(\tau_0 + \delta)}{F_n(\tau_0)} \geq \frac{c_{i_0}}{4} e^{-C_a n \delta},$$

where the last step follows from (4.5) on J . Finally,

$$n\delta \leq C(r + \sqrt{r \log(2m_S)}) \leq C(r + \log(2m_S)).$$

Absorb the fixed factors into $L = L(a)$ and use $c_{i_0} \geq \bar{c}_{i_0}$ to obtain (4.3). If $r = 0$, then $\Phi = \delta = 0$, and the same proof applies. $\square$

For the parameters of the construction, the overlap vanishes beyond a fixed angle. We record this separately because zero overlap gives more than the exponentially small bound in Lemma 4.1.

Corollary 4.2. *With $\varepsilon = 2^{-30}$ and $a = 1 + \varepsilon$, caps satisfying the hypotheses of Lemma 4.1 are disjoint whenever their axes have projective angle greater than $4\sqrt{\varepsilon}$, for all sufficiently large n .*

Proof. Each half cap has angular radius $\arccos \tau_i$. Since

$$\cos(2\sqrt{\varepsilon}) \leq 1 - 2\varepsilon + \frac{2}{3}\varepsilon^2 < 1 - \varepsilon \leq \frac{1}{1 + \varepsilon},$$

the estimate $\tau_i \geq 1/a - C_a/n$ shows that its radius is strictly less than $2\sqrt{\varepsilon}$ for large n . Every pair of signed axes is at ordinary angle at least the projective angle. The spherical triangle inequality therefore excludes a point in any pair of half caps at projective separation greater than $4\sqrt{\varepsilon}$. $\square$

5 Grouping cap conditions

We next group the cap conditions into blocks and choose a measurable set inside each common intersection. A point in this set meets every cap condition in its block. We need a lower bound on the product of the chosen measures and control of their mutual overlaps. Together these bounds will control the self-overlap of the Poisson witnesses in Section 6. The merger construction and the weighted row estimate follow Li–Liu [8, Proposition 9.1 and Lemmas 9.4–9.5]. We give the argument for arbitrary indexed lists, retaining repeated members.

Proposition 5.1 (Block estimates). *Fix constants $C_0 \geq 1$, $c_0 > 0$, and $L \geq 1$. There are constants $B_0, K, C \geq 1$ and n_1 , depending only on these three constants, with the following property. Let $n \geq n_1$, let (X, μ) be an atomless finite measure space, and let $(C_i)_{i \in I}$ be an indexed list of measurable sets with $0 < c_i := \mu(C_i) < \infty$ and $1 \leq m := |I| \leq n^2$. Set*

$$\bar{c}_i = \min(c_i, 1/2), \quad s_i = -\log \bar{c}_i,$$

and let d be a pseudometric on I . For nonempty $S \subseteq I$, write

$$r(S) = \min_{j \in S} \max_{i \in S} d(i, j)^2,$$

and choose an index $i(S) \in S$ minimizing the uncapped measure c_i . Assume that

$$\frac{\mu(C_i \cap C_j)}{\sqrt{\bar{c}_i \bar{c}_j}} \leq C_0 e^{-c_0 d(i, j)^2} \quad (i, j \in I), \quad (5.1)$$

and that, whenever $r(S) \leq B_0 \log n$,

$$\mu\left(\bigcap_{i \in S} C_i\right) \geq e^{-L(1+r(S)+\log |S|)} \bar{c}_{i(S)}. \quad (5.2)$$

Then there is a partition $\mathcal{P}$ of I and measurable sets $E_S \subseteq \bigcap_{i \in S} C_i$, $S \in \mathcal{P}$, of measures $\lambda_S = \mu(E_S)$ such that

$$n^{-K} \bar{c}_{i(S)} \leq \lambda_S \leq \frac{1}{2}. \quad (5.3)$$

Writing $q_S = -\log \lambda_S$ and

$$w_{ST} = \frac{\mu(E_S \cap E_T)}{\sqrt{\lambda_S(1+\lambda_S)\lambda_T(1+\lambda_T)}},$$

we have

$$\sum_{S \in \mathcal{P}} q_S \leq \sum_{i \in I} s_i + \frac{m}{10}, \quad (5.4)$$

$$\sum_{S \in \mathcal{P}} \log(1 + \lambda_S^{-1}) \leq C \left(m + \sum_{i \in I} s_i \right), \quad (5.5)$$

$$\sum_{S \in \mathcal{P}} \log \left(1 + \sum_{T \in \mathcal{P} \setminus \{S\}} w_{ST} \right) \leq C \left(m + \sum_{i \in I} s_i \right). \quad (5.6)$$

Proof. Set $\zeta = 1/10$ and $R_* = B_0 \log n$; we will choose B_0 after the constants governing the row estimate. Define

$$\ell(S) = \begin{cases} 0, & |S| = 1, \\ L(1 + r(S) + \log |S|), & |S| \geq 2. \end{cases}$$

For each final block S , we will choose a common subset of intensity $e^{-\ell(S)}\bar{c}_{i(S)}$. Thus $\ell(S)$ is the logarithmic loss assigned to that block. The merger condition below keeps the total loss at most ζm , while terminality will control the number of nearby blocks with small loss.

Constructing the partition. Start with the singleton partition. Whenever a subfamily of at least two current blocks has union W satisfying

$$r(W) \leq R_*, \quad \ell(W) \leq \zeta|W|, \quad (5.7)$$

replace that entire subfamily by W . The subfamily may have any size. Each merger decreases the number of blocks, so this procedure terminates after at most $m - 1$ mergers. Every resulting block satisfies

$$r(S) \leq R_*, \quad \ell(S) \leq \zeta|S|, \quad (5.8)$$

and no subfamily of at least two terminal blocks satisfies (5.7).

The anchor $i(S)$ need not attain the minimum defining $r(S)$. Applying the triangle inequality through an index that does attain it gives

$$d(i, i(S)) \leq 2\sqrt{r(S)} \leq 2\sqrt{\ell(S)/L} \quad (i \in S). \quad (5.9)$$

For singletons this holds with both sides zero. By (5.2) and atomlessness, choose $E_S \subseteq \bigcap_{i \in S} C_i$ of exactly the measure

$$\lambda_S = e^{-\ell(S)}\bar{c}_{i(S)}. \quad (5.10)$$

For singletons this choice is possible because $\bar{c}_i \leq c_i$. Thus $q_S = s_{i(S)} + \ell(S) \geq \log 2$. Different blocks have different anchor labels, and hence

$$\sum_S q_S \leq \sum_i s_i + \sum_S \zeta|S| = \sum_i s_i + \zeta m.$$

This proves (5.4). For $n \geq 3$, (5.8) and $m \leq n^2$ give

$$\ell(S) \leq L(1 + R_* + \log m) \leq L(B_0 + 3) \log n.$$

Consequently (5.3) holds with $K = L(B_0 + 3)$. Also

$$\log(1 + \lambda_S^{-1}) = q_S + \log(1 + e^{-q_S}) \leq q_S + \log(3/2) \leq \left(1 + \frac{\log(3/2)}{\log 2}\right) q_S,$$

which proves (5.5). It remains to control the overlap rows.

What terminality implies. We claim that a constant $C_{\text{ct}} = C_{\text{ct}}(L, \zeta)$ bounds the number of blocks satisfying

$$d(z, i(S))^2 \leq t, \quad \ell(S) \leq u \quad (5.11)$$

by $C_{\text{ct}}(1 + t + u)$, provided that $z \in I$, $t, u \geq 0$, and

$$8t + 8u/L \leq R_*. \quad (5.12)$$

This is a count of blocks, including blocks with coinciding anchors in pseudometric distance.

Let h be the number of such blocks and let W be their union, of size N . If $h \leq 1$ there is nothing to prove. Otherwise choose one of their anchors, say $i(S_0)$. For every $i \in S \subseteq W$, (5.9) gives

$$d(i(S_0), i) \leq d(i(S_0), z) + d(z, i(S)) + d(i(S), i) \leq 2\sqrt{t} + 2\sqrt{u/L}.$$

As $i(S_0) \in W$, it follows that

$$r(W) \leq 8t + 8u/L =: D_* \leq R_*.$$

Choose a fixed integer $N_0 \geq 2$ so large that $L \log N \leq \zeta N/2$ for all $N \geq N_0$. If

$$h \geq N_0 \quad \text{and} \quad h \geq \frac{2L}{\zeta}(1 + D_*),$$

then $N \geq h$ implies

$$\ell(W) \leq L(1 + D_* + \log N) \leq \zeta N.$$

The whole subfamily could then be merged, contradicting terminality. Since $1 + D_* \leq 8(1 + t + u)$, the claim follows, for example with $C_{\text{ct}} = 1 + N_0 + 16L/\zeta$. Here the logarithm being absorbed is $\log N$, the logarithm of the number of original labels in the union; the block sizes need not be comparable.

Estimating a row. The elementary intersection bound and (5.1) give, respectively,

$$w_{ST} \leq e^{-|q_S - q_T|/2} \leq 1, \tag{5.13}$$

$$w_{ST} \leq C_0 \exp \left\{ \frac{\ell(S) + \ell(T)}{2} - c_0 d(i(S), i(T))^2 \right\}. \tag{5.14}$$

For the first inequality, use $\mu(E_S \cap E_T) \leq \min(\lambda_S, \lambda_T)$ and discard the factors $1 + \lambda_S, 1 + \lambda_T$ in the denominator. For the second, use $E_S \subseteq C_{i(S)}$ and (5.10).

Fix S . For $k \geq 0$, put in the k th bin those $T \neq S$ for which $e^{-(k+1)} < w_{ST} \leq e^{-k}$; zero weights can be omitted. Writing $Q = q_S + k + 1 \geq 1$, the two bounds above imply

$$\ell(T) \leq q_T < q_S + 2(k + 1) \leq 2Q, \quad d(i(S), i(T))^2 \leq A_1 Q, \tag{5.15}$$

where $A_1 = (\log C_0 + 2)/c_0$. Indeed, taking logarithms in (5.14) gives

$$c_0 d(i(S), i(T))^2 < \log C_0 + \frac{q_S + q_T}{2} + k + 1 \leq \log C_0 + q_S + 2(k + 1) \leq (\log C_0 + 2)Q.$$

Now set

$$A_2 = \max\{1, 8A_1 + 16/L\}, \quad \delta = A_2^{-1}.$$

If $Q \leq \delta R_*$, apply the terminality count with $z = i(S)$, $t = A_1 Q$, and $u = 2Q$. The number of blocks in the k th bin is at most

$$C_{\text{ct}}(1 + A_1 Q + 2Q) \leq C_1(q_S + k + 1), \tag{5.16}$$

where $C_1 = C_{\text{ct}}(A_1 + 3)$. These constants are independent of B_0 . Choose $B_0 \geq \max\{1, 6/\delta\}$, and increase n_1 so that $R_* \geq 4/\delta$ for $n \geq n_1$.

First suppose $q_S \leq \delta R_*/4$, and put $J = \lfloor \delta R_*/2 \rfloor$. For $0 \leq k \leq J$ we have $q_S + k + 1 \leq \delta R_*$, so (5.16) bounds the total weight in these bins by

$$C_1 \sum_{k=0}^J e^{-k} (q_S + k + 1) \leq C_2(1 + q_S).$$

Every remaining weight is at most $e^{-(J+1)} \leq e^{-\delta R_*/2}$, and there are at most n^2 of them. Their sum is therefore at most

$$n^2 e^{-\delta R_*/2} = n^{2-\delta B_0/2} \leq 1.$$

Since $q_S \geq \log 2$, this proves

$$\log \left(1 + \sum_{T \neq S} w_{ST} \right) \leq C_3 q_S \quad (5.17)$$

in the present case. If $q_S > \delta R_*/4$, the simpler bound (5.13) gives

$$\log \left(1 + \sum_{T \neq S} w_{ST} \right) \leq \log(1 + n^2) \leq 3 \log n < \frac{12}{\delta B_0} q_S.$$

Thus (5.17) holds for every block after enlarging C_3 . Summing it and applying (5.4) proves (5.6). $\square$

Lemma 4.1 supplies the two hypotheses of Proposition 5.1 for the residual caps in the thin shell. Its constants C_0, c_0, L are independent of the fixed radius multiplier B_0 , so one first chooses these constants, then the value of B_0 specified in the proof, and finally the dimension threshold required by Lemma 4.1. All constructions use the indexed list alone. In particular, equal caps remain separate labels, and for any fixed list the partition and the measurable subsets may be chosen before the Poisson process is sampled.

6 Poisson witnesses and their lower tail

The blocks constructed in Proposition 5.1 give measurable sets of slab normals, one set for each block of residuals. Choosing a distinct Poisson point from each set certifies that all the residuals are cut. We now estimate the probability that none of a finite collection of targets has such a certificate. The lower-tail inequality originates in the work of Janson [6]. The continuous version and diagram method below follow Li–Liu [8, Lemma 6.4, Equation (6.16), and Lemma 9.6]. We include the finite inequality, its passage to measurable slots, and the matching estimates to specify all multiplicities and diagonal terms.

6.1 The finite inequality

The events in the finite approximation are indexed by ordered choices of cells. Different choices can use the same set of cells, so repeated underlying subsets must remain separate indices.

Lemma 6.1 (Finite indexed Janson inequality). *Let $(X_v)_{v \in V}$ be independent Bernoulli variables on a finite set V , and let $(H_i)_{i=1}^N$ be an indexed family of nonempty subsets of V , with repetitions allowed. Set*

$$I_i = \prod_{v \in H_i} X_v, \quad Y = \sum_{i=1}^N I_i, \quad m = \mathbb{E}Y, \quad \Delta^* = \sum_{\substack{1 \leq i, j \leq N \\ H_i \cap H_j \neq \emptyset}} \mathbb{E}(I_i I_j).$$

If $m > 0$, then $\Delta^* \geq m$ and

$$\mathbb{P}(Y = 0) \leq \exp \left(-\frac{m^2}{2\Delta^*} \right). \quad (6.1)$$

The dependency sum is ordered and includes the diagonal.

Proof. We first recall Harris’s correlation inequality for product Bernoulli measures [5], together with its elementary proof. Increasing functions of independent Bernoulli variables have nonnegative covariance. Indeed, induction on the number of variables and conditioning on the last variable reduce

the additional covariance term to $p(1-p)(f_1 - f_0)(g_1 - g_0) \geq 0$, where f_b, g_b are the corresponding conditional means. Thus an increasing event B and a decreasing event F satisfy

$$\mathbb{P}(B \mid F) \leq \mathbb{P}(B) \quad \text{whenever } \mathbb{P}(F) > 0. \quad (6.2)$$

Write $B_i = \{I_i = 1\}$, $p_i = \mathbb{P}(B_i)$, and $p_{ij} = \mathbb{P}(B_i \cap B_j)$. For each i , let F_i be the event that none of the earlier B_j with $H_i \cap H_j = \emptyset$ occurs, and let G_i exclude the remaining earlier events. If some earlier avoidance event has probability zero, the conclusion is immediate. Otherwise all the conditional probabilities below are defined. Since F_i uses only coordinates outside H_i , it is independent of B_i . It is also decreasing. A conditional union bound and (6.2) therefore give

$$\begin{aligned} \mathbb{P}(B_i \mid F_i \cap G_i) &\geq \mathbb{P}(B_i \cap G_i \mid F_i) \\ &\geq p_i - \sum_{\substack{j < i \\ H_i \cap H_j \neq \emptyset}} p_{ij}. \end{aligned}$$

The first inequality uses $\mathbb{P}(G_i \mid F_i) \leq 1$. Multiplying the successive avoidance probabilities and using $1 - x \leq e^{-x}$ yields

$$\mathbb{P}(Y = 0) \leq \exp(-m + \Delta_{\text{off}}/2), \quad \Delta_{\text{off}} = \sum_{\substack{i \neq j \\ H_i \cap H_j \neq \emptyset}} p_{ij}. \quad (6.3)$$

This remains valid when an intermediate lower bound above is negative.

Attach to each index i a private independent Bernoulli variable of parameter q , and require it as part of B_i . The new mean is qm , and its ordered off-diagonal sum is $q^2 \Delta_{\text{off}}$. The original zero event implies the new zero event, so (6.3) bounds its probability by $\exp(-qm + q^2 \Delta_{\text{off}}/2)$. Since $\Delta^* = m + \Delta_{\text{off}}$, choosing $q = m/\Delta^* \in (0, 1]$ proves (6.1). The proof retains every index even when two supports coincide. $\square$

Janson–Warnke [7, Equation (2)] record a stronger constant in this indexed formulation. The constant in Lemma 6.1 suffices here.

6.2 Measurable slots and matching diagrams

Let (Ω, μ) be a finite nonatomic standard Borel measure space, and let Π be a Poisson process of intensity μ . Consider a finite nonempty family of witness types indexed by $\alpha \in \mathcal{A}$. In the covering application, a type is a target point and its slots are blocks of residuals. For each α , let I_α be a finite nonempty set of labeled slots, with measurable sets $E_{\alpha i} \subseteq \Omega$ satisfying $\lambda_{\alpha i} := \mu(E_{\alpha i}) > 0$. Define its *activity* by

$$\nu_\alpha = \prod_{i \in I_\alpha} \lambda_{\alpha i},$$

and let Z_α count injective assignments of Poisson points to the slots, with the point assigned to i lying in $E_{\alpha i}$. These are ordered counts: equal slot sets do not identify their labels.

A *partial matching* between I_α and I_β is a set of pairs (i, j) with no repeated first or second coordinate. Put

$$D_{\alpha\beta} = \nu_\alpha \nu_\beta \sum_{\substack{M \text{ a partial matching} \\ M \neq \emptyset}} \prod_{(i,j) \in M} \frac{\mu(E_{\alpha i} \cap E_{\beta j})}{\lambda_{\alpha i} \lambda_{\beta j}}. \quad (6.4)$$

The Poisson factorial-moment formula gives

$$\mathbb{E}Z_\alpha = \nu_\alpha, \quad \mathbb{E}[Z_\alpha Z_\beta] = \nu_\alpha \nu_\beta + D_{\alpha\beta}.$$

For the second identity, group pairs of injective assignments by the partial matching of slots occupied by the same point. The empty matching contributes $\nu_\alpha \nu_\beta$; the nonempty matchings give exactly the terms in (6.4). Thus $D_{\alpha\beta}$ is the covariance of Z_α and Z_β .

Proposition 6.2 (Poisson lower tail). *Let Π be a Poisson process on a finite nonatomic standard Borel measure space. For each member of a finite nonempty family, take a finite nonempty labeled list of measurable slots of positive intensity. Let Z_α count their injective assignments, let ν_α be the product of their intensities, and define $D_{\alpha\beta}$ by (6.4). Then $\mathbb{E}Z_\alpha = \nu_\alpha$, and*

$$\mathbb{P}(Z_\alpha = 0 \text{ for all } \alpha \in \mathcal{A}) \leq \exp \left[-\frac{(\sum_\alpha \nu_\alpha)^2}{2 \sum_{\alpha, \beta} D_{\alpha\beta}} \right]. \quad (6.5)$$

The denominator is positive: the full identity matching in $D_{\alpha\alpha}$ contributes ν_α .

Proof. Write $\tau = \mu(\Omega) > 0$. Conditional on $N = \Pi(\Omega)$, the points are independent with law μ/τ and are almost surely distinct. If $k = |I_\alpha|$, the Poisson factorial moment $\mathbb{E}(N)_k = \tau^k$ gives

$$\mathbb{E}Z_\alpha = \mathbb{E}(N)_k \tau^{-k} \prod_i \lambda_{\alpha i} = \nu_\alpha.$$

In particular, no factorial divides the activity, even for repeated slot sets.

We apply Lemma 6.1 to cell-occupation indicators. Fix $\delta > 0$. Form the Boolean atoms generated by all the slot sets and subdivide each positive-measure atom into finitely many measurable cells of measure at most δ , using nonatomicity. After omitting null cells, this gives a finite partition $\mathcal{P}_\delta$ that respects every slot. Modifications on the common null set do not affect the Poisson process. For $C \in \mathcal{P}_\delta$, set

$$a_C = \mu(C), \quad X_C = \mathbf{1}_{\{\Pi(C) > 0\}}, \quad p_C = \mathbb{E}X_C = 1 - e^{-a_C}.$$

These variables are independent and

$$0 \leq a_C - p_C \leq \frac{1}{2}a_C^2 \leq \frac{\delta}{2}a_C. \quad (6.6)$$

For each α , retain every injective assignment of cells $f : I_\alpha \rightarrow \mathcal{P}_\delta$ with $f(i) \subseteq E_{\alpha i}$. Its indicator is $\prod_i X_{f(i)}$. Let Y_δ be the sum over these indexed assignments. Different assignments remain different indices even when their underlying cell sets agree.

The following estimate controls all cell collisions. For a fixed list $F_1, \dots, F_r$ of unions of cells, let $Q_\delta(F_1, \dots, F_r)$ be the sum of $\prod_{v=1}^r p_{C_v}$ over distinct cells $C_v \subseteq F_v$. Then

$$\begin{aligned} 0 \leq \prod_v \mu(F_v) - Q_\delta(F_1, \dots, F_r) &\leq \delta \left[\frac{r}{2} \prod_v \mu(F_v) \right. \\ &\quad \left. + \sum_{v < w} \mu(F_v \cap F_w) \prod_{u \neq v, w} \mu(F_u) \right]. \end{aligned} \quad (6.7)$$

Indeed, the unconstrained sum factors over the slots, and (6.6), followed by telescoping the product, bounds its difference from $\prod_v \mu(F_v)$ by the first term. The weight of assignments with $C_v = C_w$ is at most

$$\sum_{C \subseteq F_v \cap F_w} p_C^2 \prod_{u \neq v, w} \mu(F_u) \leq \delta \mu(F_v \cap F_w) \prod_{u \neq v, w} \mu(F_u).$$

A union bound proves (6.7). Empty products are 1, and the estimate also holds if a set has measure zero. Applying it to each slot list gives

$$m_\delta := \mathbb{E}Y_\delta \longrightarrow \sum_\alpha \nu_\alpha. \quad (6.8)$$

It remains to identify the finite dependency sum and the limiting event. For two internally injective assignments f and g , the complete set of equalities $f(i) = g(j)$ is a unique partial matching M . Their Bernoulli supports intersect precisely when $M \neq \emptyset$. For fixed α, β, M , identify the slots paired by M and leave the other slots as singletons. Assign to each resulting class its slot set, or the intersection of its two slot sets. Denote these sets by $(F_v)_v$. Requiring distinct cells for these classes excludes exactly the unintended equalities. Thus the contribution with exact matching M is $Q_\delta((F_v)_v)$, and its limit is

$$\prod_v \mu(F_v) = \nu_\alpha \nu_\beta \prod_{(i,j) \in M} \frac{\mu(E_{\alpha i} \cap E_{\beta j})}{\lambda_{\alpha i} \lambda_{\beta j}}.$$

All sums are finite. Estimate (6.7) therefore shows that the ordered dependency sum, including identical assignment pairs, satisfies

$$\Delta_\delta^* \longrightarrow \sum_{\alpha, \beta} D_{\alpha\beta}. \quad (6.9)$$

Identical assignments give the full identity matching; other full matchings caused by repeated slots remain separate terms.

Finally, couple the cell variables with the original process. The probability that any cell contains two or more points is at most

$$\frac{1}{2} \sum_C \mathbb{E}[\Pi(C)(\Pi(C) - 1)] = \frac{1}{2} \sum_C a_C^2 \leq \frac{\delta\tau}{2}.$$

Outside this event, point injections and occupied-cell injections are in bijection, simultaneously for every witness type. Hence

$$|\mathbb{P}(Y_\delta = 0) - \mathbb{P}(Z_\alpha = 0 \text{ for all } \alpha)| \leq \frac{\delta\tau}{2}.$$

Apply Lemma 6.1 and let $\delta \downarrow 0$, using (6.8) and (6.9). The identity-matching contribution makes the limiting denominator positive. This proves (6.5). $\square$

The approximation is performed for each fixed finite family. Its size and the total intensity may grow with dimension; no uniform estimate in those quantities is needed before taking the cell limit. An empty slot list has one empty injection and is treated as a deterministic certificate before applying Proposition 6.2.

6.3 Self and cross diagrams

The two budgets in Proposition 5.1 now control the self-diagram sum. The factor $1 + \lambda_i^{-1}$ pays for identifying the same slot in both copies of a witness.

Lemma 6.3 (Matching bounds). *For positive-measure slot lists and the diagrams in (6.4), the following bounds hold. For one witness type, suppress its label and set*

$$\delta_i = 1 + \lambda_i^{-1}, \quad w_{ij} = \frac{\mu(E_i \cap E_j)}{\sqrt{\lambda_i(1 + \lambda_i)\lambda_j(1 + \lambda_j)}} \quad (i \neq j).$$

Then

$$1 + \frac{D_{\alpha\alpha}}{\nu_\alpha^2} \leq \prod_i \delta_i \prod_i \left(1 + \sum_{j \neq i} w_{ij} \right). \quad (6.10)$$

For two witness types, put

$$\rho_{\alpha\beta} = \sum_{i \in I_\alpha, j \in I_\beta} \frac{\mu(E_{\alpha i} \cap E_{\beta j})}{\lambda_{\alpha i} \lambda_{\beta j}}.$$

Then

$$D_{\alpha\beta} \leq \nu_\alpha \nu_\beta (e^{\rho_{\alpha\beta}} - 1). \quad (6.11)$$

In particular, if every cross-slot intersection has measure zero, then $D_{\alpha\beta} = 0$.

Proof. Write $h_{ij} = \mu(E_i \cap E_j) / (\lambda_i \lambda_j)$. For the self sum, fix the off-diagonal edges P of a partial matching between two copies of the slot list. Regard P as a directed graph on the common label set. Every vertex has total degree $d_P(i) \in \{0, 1, 2\}$, since its in-degree and out-degree are at most one. A diagonal loop is available exactly at the vertices of degree zero. Since $h_{ii} = \lambda_i^{-1}$, summing the compatible loop choices gives

$$\left(\prod_i \delta_i \right) \left(\prod_{(i,j) \in P} h_{ij} \right) \left(\prod_{i: d_P(i) > 0} \delta_i^{-1} \right) \leq \left(\prod_i \delta_i \right) \prod_{(i,j) \in P} w_{ij}.$$

Here $\delta_i^{-1} \leq \delta_i^{-d_P(i)/2}$ at each incident vertex. This includes directed paths and cycles; Figure 2 illustrates the permitted incidences. Summing over P and dropping the requirement of distinct right endpoints bounds the remaining sum by $\prod_i (1 + \sum_{j \neq i} w_{ij})$. The empty matching contributes 1, proving (6.10).

For two lists, drop all matching restrictions and sum over nonempty subsets of cross edges. With the analogous nonnegative weights h_{ij} , the result is at most $\prod_{i,j} (1 + h_{ij}) - 1 \leq e^{\sum_{i,j} h_{ij}} - 1$. This proves (6.11). $\square$

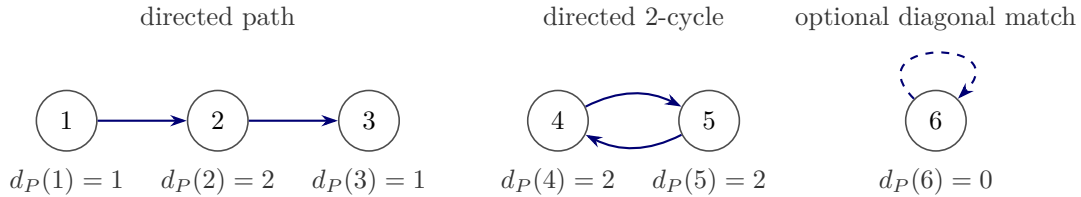


Figure 2: A self-matching after identifying the two copies of the slot labels. The solid edges form the off-diagonal matching $P = \{(1, 2), (2, 3), (4, 5), (5, 4)\}$; each label has at most one incoming and one outgoing edge. The degree $d_P(i)$ counts both. A diagonal match is available only at an isolated label, as illustrated by the optional dashed loop at 6; even the degree-one path endpoints cannot receive a diagonal match.

6.4 Application to residual caps

We return to the fixed parameters and the spherical intensity μ of Section 2. The following consequence collects the inputs needed for the target selection in the next section.

Corollary 6.4 (Witnesses for residual lists). *Fix $A_0 > 0$. There are constants $A, C_*, H > 0$ such that, for all sufficiently large n , the following holds. Let $R \geq 1$, and let $(x_{\alpha i})_{i=1}^{m_\alpha}$, $\alpha \in \mathcal{A}$, be any finite nonempty family of deterministic lists satisfying*

$$1 \leq m_\alpha \leq n^2, \quad r_- - 3\eta \leq |x_{\alpha i}| \leq b + 3\eta.$$

Define

$$C_{\alpha i} = \{u \in S^{n-1} : |u \cdot x_{\alpha i}| > 1 + 3\eta\}, \quad s_{\alpha i} = -\log \min\{\mu(C_{\alpha i}), 1/2\},$$

and suppose $m_\alpha + \sum_i s_{\alpha i} \leq A_0 R$. For each list there is a nonempty block partition and measurable slots $E_{\alpha S} \subseteq \bigcap_{i \in S} C_{\alpha i}$, chosen independently of Π , with positive intensities at most $1/2$. Their activities and diagrams satisfy

$$e^{-AR} \leq \nu_\alpha \leq 1, \quad D_{\alpha\alpha} \leq \nu_\alpha^2 e^{C_* R}. \quad (6.12)$$

For distinct α, β , if all cross-residual angles exceed $H\sqrt{\log n/n}$, then

$$D_{\alpha\beta} \leq n^{-3} \nu_\alpha \nu_\beta. \quad (6.13)$$

If all those angles exceed $4\sqrt{\varepsilon}$, then $D_{\alpha\beta} = 0$. A positive injection count for a list implies that every cap in that list is hit by Π .

Proof. Apply Lemma 4.1 with shell parameter 10 and perturbation bound 3. Its affinity and common-intersection estimates supply the hypotheses of Proposition 5.1; the common-intersection constant is available before the radius multiplier in that proposition is fixed. Apply the proposition separately to each list, making all choices before sampling Π . The budget (5.5) gives

$$-\log \nu_\alpha \leq \sum_S \log(1 + \lambda_{\alpha S}^{-1}) \leq AR.$$

Together with (5.6) and (6.10), it also gives the self bound in (6.12), after choosing C_* .

To check the cross estimate, write $\bar{c}_{\alpha i} = \min\{\mu(C_{\alpha i}), 1/2\}$ and let $i(S)$ be the anchor of a block. Lemma 4.1 and Proposition 5.1 provide fixed exponents $P_0, P_1 > 0$ such that

$$\bar{c}_{\alpha i} \geq n^{-P_0}, \quad \lambda_{\alpha S} \geq n^{-P_1} \bar{c}_{\alpha, i(S)}, \quad E_{\alpha S} \subseteq C_{\alpha, i(S)}.$$

Apply the affinity estimate to each two-element list of anchor residuals; this does not require the union of the two full lists to have at most n^2 entries. It gives

$$\frac{\mu(E_{\alpha S} \cap E_{\beta T})}{\lambda_{\alpha S} \lambda_{\beta T}} \leq C n^{2P_1 + P_0} e^{-cn \angle(x_{\alpha, i(S)}, x_{\beta, i(T)})^2}.$$

There are at most n^4 block pairs. Choose H after these constants, so that $cH^2 > 9 + 2P_1 + P_0$. For sufficiently large n , the angular hypothesis gives $\rho_{\alpha\beta} \leq n^{-4}$. Equation (6.11) now gives $D_{\alpha\beta} \leq 2n^{-4} \nu_\alpha \nu_\beta$, which implies (6.13). At cross angles greater than $4\sqrt{\varepsilon}$, Corollary 4.2 makes every cross-slot intersection empty, and the diagram is zero.

Finally, the point assigned to a block lies in every cap belonging to that block. A positive injection count thus hits every residual cap. $\square$

For a finite center list (p_i) and a target y , take all residuals $x_i = y - p_i$ with $|x_i| \leq b + 3\eta$. If this list satisfies the hypotheses of Corollary 6.4, a positive witness excludes each such residual from K^+ by a strict slab inequality. All remaining residuals are excluded by its outer ball. Consequently

$$Z_y > 0 \implies y \notin \bigcup_i (p_i + K^+). \quad (6.14)$$

If the relevant list is empty, the target is already a hole. Proposition 6.2 supplies an exponential bound on the probability that every chosen target lacks a witness. A second-moment bound alone would not give the decay required for the finite-pattern union in the final argument.

7 Many targets for each finite pattern

We now fix one of the rounded center lists supplied by Proposition 3.1. Our goal is to bound the probability that this list covers the window B_D by translates of K^+ . The good subset of the window contains many points whose residual vectors are separated at a small angular scale. Only a small fraction of pairs need fail a larger, fixed separation. The former property controls every cross diagram; the latter makes most cross diagrams vanish.

7.1 A deterministic choice of targets

The following volume estimate controls projective angles, including both possible orientations of a residual vector.

Lemma 7.1 (Cylinder estimate). *Let $n \geq 2$, let $D, L > 0$, and let $G \subset B_D$ be measurable with $|G| \geq \omega_n D^n / 4$. Let Y, Z be independent uniform points of G , and let $p, q \in \mathbb{R}^n$. For $0 < \xi < \pi/2$,*

$$\begin{aligned} \mathbb{P}(0 < |Y - p| \leq L, 0 < |Z - q| \leq L, \angle(Y - p, Z - q) \leq \xi) \\ \leq 8 \frac{\omega_{n-1}}{\omega_n} \left(\frac{L \sin \xi}{D} \right)^{n-1} \leq 8\sqrt{n} \left(\frac{L \sin \xi}{D} \right)^{n-1}. \end{aligned} \quad (7.1)$$

Proof. Condition on $Y = y$ with $0 < |y - p| \leq L$, and write $u = (y - p)/|y - p|$. If the event in the statement occurs, the component of $Z - q$ perpendicular to u has length at most $L \sin \xi$. Both orientations allowed by the projective angle therefore lie in the same cylinder around the affine line $q + \mathbb{R}u$. The projection of B_D onto the direction u has length $2D$. Every perpendicular slice of the cylinder has $(n-1)$ -dimensional volume at most $\omega_{n-1}(L \sin \xi)^{n-1}$. Thus the allowed points Z have volume at most $2D\omega_{n-1}(L \sin \xi)^{n-1}$. Divide by $|G|$ and then average over Y .

To obtain the last inequality, use the slicing identity

$$\frac{\omega_n}{\omega_{n-1}} = \int_{-1}^1 (1 - t^2)^{(n-1)/2} dt \geq \frac{2 \exp(-1/2)}{\sqrt{n}}.$$

Indeed, restrict the integral to $|t| \leq n^{-1/2}$ and use $\log(1 - 1/n) \geq -1/(n-1)$. In particular, $\omega_{n-1}/\omega_n \leq \sqrt{n}$. $\square$

For a finite center list $P = (p_1, \dots, p_N)$ and a point y different from its entries, write

$$\mathcal{R}_P(y) = (y - p_i : |y - p_i| \leq b + 3\eta)$$

for its labeled list of relevant residuals. For $\xi > 0$, call a pair (y, z) ξ -bad if some residual in $\mathcal{R}_P(y)$ and some residual in $\mathcal{R}_P(z)$ have projective angle at most ξ . This definition is symmetric in y, z and retains repeated center labels. Recall γ from (2.3), and put

$$M_n = \left\lfloor \exp \left(\frac{n \log n}{16} \right) \right\rfloor. \quad (7.2)$$

Lemma 7.2 (Separated targets). *Fix $H > 0$ and $k_0 > 0$. For all sufficiently large n , let $P = (p_1, \dots, p_N)$ be a deterministic list satisfying*

$$N \leq n^{k_0} \exp(\lambda_0 n),$$

and let $G \subset B_D \setminus \{p_1, \dots, p_N\}$ be measurable with $|G| \geq \omega_n D^n / 4$. There are M_n distinct points $y_1, \dots, y_{M_n} \in G$ such that:

- (i) no pair with distinct indices is $H\sqrt{\log n/n}$ -bad;
- (ii) at most a fraction $10f_n$ of the unordered pairs with distinct indices is $4\sqrt{\varepsilon}$ -bad, where

$$f_n = 8\sqrt{n} n^{2k_0} \exp(2\lambda_0 n) \left(\frac{8\sqrt{\varepsilon}}{D} \right)^{n-1} = \exp(-\gamma n + O(\log n)). \quad (7.3)$$

The constants and the dimension threshold are uniform in P and G . The selected points can be fixed using only P and G .

Proof. Sample M_n points independently and uniformly from G . For independent sampled points Y, Z , take the union in Lemma 7.1 over all N^2 ordered pairs of center labels. Since $b + 3\eta \leq 2$ for sufficiently large n , this gives

$$\mathbb{P}((Y, Z) \text{ is } \xi\text{-bad}) \leq 8\sqrt{n} N^2 \left(\frac{2 \sin \xi}{D} \right)^{n-1}. \quad (7.4)$$

All relevant labels are included even though the relevant lists vary with Y and Z . Repeated centers only repeat events in this union.

Set $t_n = H\sqrt{\log n/n}$. Multiply the right side of (7.4), with $\sin \xi \leq t_n$, by $\binom{M_n}{2}$. The logarithm of the resulting bound is at most

$$-\frac{3}{8}n \log n + \frac{1}{2}n \log \log n + O_{H,k_0}(n),$$

which tends to $-\infty$. Thus the probability that any pair of sampled points is t_n -bad tends to zero, uniformly in the pattern. For $\xi = 4\sqrt{\varepsilon}$, the inequality $\sin \xi \leq 4\sqrt{\varepsilon}$ instead bounds (7.4) by f_n . The expected fraction of these bad unordered pairs is at most f_n . Markov's inequality shows that this fraction exceeds $10f_n$ with probability at most $1/10$. Consequently both conclusions hold with probability at least $9/10 - o(1) > 0$. The sampled points are distinct almost surely because uniform measure on G has no atoms. Fix a successful list. No independence between the two separation properties is needed. $\square$

7.2 From targets to a small covering probability

We apply the preceding selection before exposing the Poisson process. For each selected target, Corollary 6.4 supplies sets of slab normals whose simultaneous occupancy certifies that the target is uncovered. Binning their activities will let us combine many certificates without losing their angular separation.

Proposition 7.3 (Failure probability for one pattern). *There are constants $c_0, C_{\text{tail}} > 0$ and n_0 such that, whenever $n \geq n_0$ and $1 \leq R \leq c_0 n \log n$, every retained pattern $P \in \mathcal{F}_n$ of Proposition 3.1 satisfies*

$$\mathbb{P} \left(B_D \subseteq \bigcup_{p \in P} (p + K^+) \right) \leq \exp [-\exp(\gamma n - C_{\text{tail}} \log n)]. \quad (7.5)$$

The constants are independent of the pattern, its repetitions, and R .

The covering event is measurable. Indeed, the finite union of translates of K^+ is closed, so it covers B_D exactly when it contains a fixed countable dense subset of B_D . At each such point, membership is a finite union of ball conditions and Poisson cap-avoidance events.

Proof. Fix $P \in \mathcal{F}_n$ and its deterministic good set G_P . At each $y \in G_P$, Proposition 3.1 gives

$$r_- - 2\eta \leq |x_i| \leq b + 3\eta, \quad m_y \leq 2C_{\text{loc}}R, \quad \sum_{i=1}^{m_y} w_a(|x_i|) \leq 2C_{\text{loc}}R \quad (7.6)$$

for its relevant residuals $x_i = y - p_i$. If this list is empty at any point of G_P , that point lies in none of the balls $p + (b + 3\eta)B$, so the probability in (7.5) is zero. We may assume all these lists are nonempty. For the range of R in the proposition, $m_y \leq n^2$ once n is sufficiently large.

For each relevant residual, define the cutting cap

$$C_i(y) = \{u \in S^{n-1} : |u \cdot x_i| > 1 + 3\eta\}.$$

Lemma 4.1, applied with radial loss $10 \log n/n$ and a fixed perturbation bound, gives

$$s_i = -\log \min\{\mu(C_i(y)), 1/2\} \leq C_a w_a(|x_i|).$$

Consequently

$$m_y + \sum_{i=1}^{m_y} s_i \leq A_0 R, \quad A_0 = 2C_{\text{loc}}(1 + C_a). \quad (7.7)$$

All assumptions of Corollary 6.4 therefore hold uniformly in y and P . Fix its constants A, C_* and H for this value of A_0 .

The list-length estimate in Proposition 3.1 gives $N \leq n^{k_0} \exp(\lambda_0 n)$ for a fixed k_0 , uniformly in the present range of R . Apply Lemma 7.2 to G_P with the just-fixed H . At each of the resulting $M = M_n$ targets, choose the block partition and measurable witness sets from Corollary 6.4. Denote its injection count by Z_y , its activity by ν_y , and the diagram sums by D_{yz} . The corollary gives

$$\exp(-AR) \leq \nu_y \leq 1, \quad D_{yy} \leq \nu_y^2 \exp(C_* R). \quad (7.8)$$

Every pair of distinct selected targets satisfies $D_{yz} \leq n^{-3} \nu_y \nu_z$. Moreover, $D_{yz} = 0$ unless the pair is $4\sqrt{\varepsilon}$ -bad. Our use of “at most” in the definition of badness makes both angular interfaces strict on its complement.

All choices so far depend only on P and its targets. In particular, the witness sets are chosen before the Poisson process; their activities are deterministic intensities. There is no need to choose witness sets measurably at every point of G_P , since only the finite selected list is used. If $Z_y > 0$, every relevant residual is cut by a slab of the process, and the other centers are too far from y to contain it in a translate of K^+ . Thus $Z_y > 0$ certifies a hole.

Divide the possible values of $-\log \nu_y$ in $[0, AR]$ into intervals of length $\log 2$. At most

$$Q = 1 + \left\lceil \frac{AR}{\log 2} \right\rceil$$

bins suffice. A largest bin has $m \geq M/Q$ targets and, for some $\nu > 0$, all its activities belong to $[\nu, 2\nu]$. There are at most $10f_n \binom{M}{2}$ bad unordered pairs in the full list. Hence the chosen bin contains at most

$$10f_n M(M-1) \leq 10f_n Q^2 m^2 \quad (7.9)$$

bad ordered pairs with distinct indices. This bound allows the bin to contain every bad pair; no independence between bin membership and angles is asserted.

Apply Proposition 6.2 to the witnesses in this bin. Its numerator is at least $m^2 \nu^2$. By (7.8) and (7.9), its denominator before the factor 2 is at most

$$\sum_{y,z} D_{yz} \leq 4m\nu^2 \exp(C_* R) + 40m^2 \nu^2 n^{-3} Q^2 f_n.$$

The first term counts diagonal pairs. Among distinct pairs, only the coarse bad pairs contribute. Coverage forces $Z_y = 0$ at every target of the bin, so

$$\mathbb{P}\left(B_D \subseteq \bigcup_{p \in P} (p + K^+)\right) \leq \exp\left[-\frac{1}{8(\exp(C_*R)/m + 10n^{-3}Q^2f_n)}\right]. \quad (7.10)$$

Now choose

$$c_0 = \frac{1}{64 \max\{C_*, 1\}}.$$

Uniformly for $1 \leq R \leq c_0 n \log n$, we have $\log Q = O(\log n)$ and, for sufficiently large n ,

$$m \exp(-C_*R) \geq \frac{M_n}{Q} \exp(-C_*R) \geq \exp\left(\frac{n \log n}{32}\right).$$

The first term inside the parentheses in (7.10) is therefore at most $\exp(-n \log n/32)$. By (7.3), the second is $\exp(-\gamma n + O(\log n))$. Since γ is fixed, their sum is at most $\exp(-\gamma n + O(\log n))$; absorbing the fixed factor 8 gives (7.5).

The order of the choices is fixed throughout: the geometric parameters come first, then the cap and block constants and H , then c_0 , and finally the dimension threshold. Every estimate is uniform over the retained finite class. In particular, the auxiliary target selection introduces no dependence on the realization of Π . $\square$

8 Completion of the proof

We now combine the estimates and specify their order of use. The numerical parameters (2.1) are already fixed. Apply Lemma 4.1 with the shell and perturbation constants required by Proposition 3.1. Its affinity and common-intersection constants determine the constants in Proposition 5.1, including its radius cutoff. The radial cost estimate gives $A_0 = 2C_{\text{loc}}(1 + C_a)$ in the residual budget. For this value of A_0 , Corollary 6.4 fixes the cross-angle constant H and the self-diagram coefficient C_* . Choose a positive rational number

$$c \leq \frac{1}{64 \max(C_*, 1)}. \quad (8.1)$$

Every constant in this order is independent of dimension. Increase n_0 so that all preceding estimates hold when $n \geq n_0$ and $R = cn \log n$. In particular, $R \geq 1$, $R \leq n^2$, and every relevant cap list has at most n^2 entries.

Proof of Theorem 1.1. Use the deterministic class $\mathcal{F}_n$ from Proposition 3.1, with $R = cn \log n$. That proposition and Proposition 7.3 give

$$\begin{aligned} & \mathbb{P}\left\{\text{some } P \in \mathcal{F}_n \text{ covers } B_D \text{ by translates of } K^+\right\} \\ & \leq \exp(\exp(\lambda_0 n + O(\log n)) - \exp(\gamma n - O(\log n))) = o(1). \end{aligned}$$

The error bounds are uniform over this finite class. Since $\gamma > \lambda_0$, the negative term dominates. Lemma 2.1 therefore guarantees a realization for which $|K| > 2v\omega_n a^n$ and no retained pattern covers B_D by translates of K^+ .

Suppose that $K + \Lambda + T = \mathbb{R}^n$ with $|T|/\det \Lambda \leq \rho_0$. Proposition 3.1 would give a retained pattern covering B_D by K^+ , a contradiction. Every covering thus has center intensity greater than ρ_0 . Taking the infimum of its density yields

$$\theta_T(K) = \theta_{\text{per}}(K) \geq \rho_0 |K| > R = cn \log n,$$

where Proposition A.1 supplies the equality. The realization is a centrally symmetric convex body, as observed after (2.6). Choose one for each $n \geq n_0$. $\square$

The exclusion is simultaneous over all full-rank lattices and all finite coset lists. The lattice and the centers may depend on the chosen body: localization still places their rounded list in the class fixed before sampling. The periodization equivalence transfers this exclusion to unrestricted locally finite coverings without changing the body or the constant.

8.1 The optimal order for translative and lattice coverings

For a convex body K , define its single-lattice covering density by

$$\theta_L(K) = \inf_{\Lambda: K+\Lambda=\mathbb{R}^n} \frac{\text{vol}(K)}{\det \Lambda}.$$

Every lattice covering is periodic, so $\theta_T(K) = \theta_{\text{per}}(K) \leq \theta_L(K)$. Let $\mathcal{K}^n$ denote the convex bodies in $\mathbb{R}^n$ and $\mathcal{K}_{\text{sym}}^n$ their centrally symmetric members.

Corollary 8.1. *As $n \rightarrow \infty$, each of the four quantities*

$$\begin{aligned} \sup_{K \in \mathcal{K}^n} \theta_T(K), & \quad \sup_{K \in \mathcal{K}_{\text{sym}}^n} \theta_T(K), \\ \sup_{K \in \mathcal{K}^n} \theta_L(K), & \quad \sup_{K \in \mathcal{K}_{\text{sym}}^n} \theta_L(K) \end{aligned}$$

has order $n \log n$, with absolute comparison constants.

Proof. Theorem 1.1 gives the lower bound already among centrally symmetric bodies, and $\theta_T \leq \theta_L$ transfers it to lattice coverings. Rogers' bound (1.3) supplies the translative upper bound. The companion single-lattice theorem [10, Theorem 1.1] gives $\theta_L(K) \leq Cn \log n$ for every convex body and every $n \geq 2$, which supplies the two lattice upper bounds. $\square$

9 Rational polytopes and normal lists

The strict volume margin in the proof gives a polyhedral consequence of Theorem 1.1 without changing its constant.

Corollary 9.1. *With the same c and n_0 as in Theorem 1.1, for every $n \geq n_0$ there is a centrally symmetric polytope $P_n \subset \mathbb{R}^n$ with rational vertices such that*

$$\theta_T(P_n) > cn \log n.$$

Proof. Fix the successful body K in the proof of Theorem 1.1, and put $R = cn \log n$. It satisfies $|K| > 2v\omega_n a^n$, and no periodic covering by K has center intensity at most $\rho_0 = R/(2v\omega_n a^n)$.

Enumerate the rational points of $\text{int } K$ as $q_1, q_2, \dots$ and set

$$P_j = \text{conv}\{\pm q_1, \dots, \pm q_j\}.$$

These are centrally symmetric rational polytopes contained in $\text{int } K$, and they increase with j . Their union is $\text{int } K$: every interior point lies inside a sufficiently small simplex with rational vertices in $\text{int } K$, and all its vertices eventually occur in the enumeration. Continuity of volume from below,

together with the fact that the boundary of a convex body has measure zero, gives $|P_j| \rightarrow |K|$. Choose j so that $|P_j| > 2v\omega_n a^n$; in particular, this P_j has nonempty interior.

Every periodic covering by P_j is a covering by K , because $P_j \subset K$. Its center intensity therefore exceeds ρ_0 . Taking the infimum of the covering densities and using Proposition A.1 yields

$$\theta_T(P_j) = \theta_{\text{per}}(P_j) \geq \rho_0 |P_j| > R.$$

This argument uses both inclusion and the retained volume margin. $\square$

A different finite description retains the spherical truncation bB and makes all slab normals rational unit vectors. These bodies need not be polytopes.

Proposition 9.2. *The bodies in Theorem 1.1 can be chosen in the form*

$$K = bB \cap \bigcap_{j=1}^m \{x : |u_j \cdot x| \leq 1\}, \quad u_j \in \mathbb{Q}^n \cap S^{n-1},$$

with $m < \infty$.

Proof. Fix a successful finite normal list $U = (u_1, \dots, u_m)$ from the proof of Theorem 1.1. The class $\mathcal{F}_n$ is finite and independent of U . For each $P \in \mathcal{F}_n$, select a point $y_P \in B_D$ outside all of its translates of $K^+(U)$. For every center p in that list, either $|y_P - p| > b + 3\eta$, or one of the normals satisfies

$$|u_j \cdot (y_P - p)| > 1 + 3\eta.$$

Each required inequality is strict. There are finitely many patterns, centers, and chosen certificates, so all these inequalities persist under a sufficiently small simultaneous perturbation of the normals. Empty patterns need no certificates.

The strict volume test persists as well. If U' is a list of unit normals with $|u_j - u'_j| \leq \delta$ for every j , then

$$(1 + b\delta)^{-1} K(U) \subset K(U') \subset (1 + b\delta) K(U). \quad (9.1)$$

Indeed, if $x \in K(U)$, then $|x| \leq b$ and $|u'_j \cdot x| \leq 1 + b\delta$; scaling by $(1 + b\delta)^{-1}$ proves the first inclusion. Interchanging U and U' proves the second. Volumes therefore converge as $\delta \rightarrow 0$.

Rational unit vectors are dense in S^{n-1} for $n \geq 2$, by the stereographic parametrization

$$t \mapsto \frac{(2t, 1 - |t|^2)}{1 + |t|^2}, \quad t \in \mathbb{Q}^{n-1},$$

and approximation of its omitted pole. Approximate all normals closely enough to preserve the holes and the strict volume test. The localization argument applies to every finite normal list, so the resulting rational list yields the same lower bound. $\square$

One can specify a choice exactly by ordering the finite rational normal lists. Write each coordinate in reduced form with positive denominator, and let its height be the maximum of the absolute numerator and denominator. Order lists by their length plus maximum coordinate height (height 1 for the empty list), breaking each finite tie lexicographically. The lists satisfying the strict volume test and every finite-pattern noncoverage test form a nonempty set, so they have a first member. This specifies a finite mathematical object; it does not assert an algorithm for deciding the volume comparisons or an effective bound for the first admissible dimension.

A Covering density and periodization

The proof constructs bodies that exclude every finite union of lattice cosets. We verify that this gives the same lower bound for unrestricted translative coverings, with a precise density convention. This standard periodization principle is recorded in Prosanov [11, Theorem 2.1], with attribution to Rogers. We give the finite-window argument to make its boundary and limiting conventions explicit.

Fix a convex body $K \subset \mathbb{R}^n$ and write $V = \text{vol}(K)$. Use the definitions of θ_T and θ_{per} from Section 1. For a locally finite covering center set X , the centered cube lower density replaces the lim sup in (1.1) by lim inf. The centered ball versions use B_R and $\text{vol}(B_R)$ in place of Q_R and $(2R)^n$.

Proposition A.1 (Periodization). *For every convex body $K \subset \mathbb{R}^n$,*

$$\theta_T(K) = \theta_{\text{per}}(K).$$

The same infimum is obtained by replacing the centered cube upper density by centered cube lower density or centered Euclidean-ball lower or upper density. These are equalities of infima over coverings; the corresponding statistics of one covering need not agree.

Proof. Choose $r > 0$ with $K \subset Q_r$, and let X be a locally finite covering center set. For $a \in \mathbb{R}^n$ and $L > 0$, the finite set

$$S = X \cap (a + Q_{L+r})$$

covers $a + Q_L$ by translates of K . Indeed, if $y \in a + Q_L$ and $y = k + x$ with $k \in K$ and $x \in X$, then $x = y - k \in a + Q_{L+r}$. This holds on the boundary as well. Repeating S with periods $2L\mathbb{Z}^n$ covers all of $\mathbb{R}^n$ because the cubes $a + Q_L + 2Lz$ cover it. Reducing S modulo this lattice gives

$$\theta_{\text{per}}(K) \leq \frac{V \#(X \cap (a + Q_{L+r}))}{(2L)^n}. \quad (\text{A.1})$$

Writing $R = L + r$, we obtain, for every a and every $R > r$,

$$\frac{V \#(X \cap (a + Q_R))}{(2R)^n} \geq \theta_{\text{per}}(K) (1 - r/R)^n. \quad (\text{A.2})$$

This proves the required lower bound for both centered cube conventions.

For the ball conventions, fix $L > 0$ and put $\beta = \sqrt{n}(L + r)$. Integrate (A.1) over $a \in z + B_{R-\beta}$, where $R > \beta$. Any counted center belongs to $z + B_R$, and each such center is counted on a set of a of volume at most $(2(L + r))^n$. Tonelli's theorem therefore gives

$$\frac{\theta_{\text{per}}(K)(2L)^n}{V} \text{vol}(B_{R-\beta}) \leq \int_{z+B_{R-\beta}} \#(X \cap (a + Q_{L+r})) da \leq (2(L + r))^n \#(X \cap (z + B_R)).$$

Consequently, uniformly in z ,

$$\frac{V \#(X \cap (z + B_R))}{\text{vol}(B_R)} \geq \theta_{\text{per}}(K) \left(\frac{L}{L + r} \right)^n (1 - \beta/R)^n.$$

First let $R \rightarrow \infty$ with L fixed, and then let $L \rightarrow \infty$. This proves the lower bound for the ball lower densities and hence also for the upper densities. No comparison of a ball with just one containing or contained cube is needed.

For the reverse inequality, consider a periodic center set $X = \Lambda + T$ with distinct cosets. Let F be a bounded half-open fundamental parallelepiped of Λ and choose $d > 0$ with $F \subset Q_d \cap B_d$. For each fixed $t \in T$, the cells $t + \lambda + F$ partition space up to null boundaries. For $R > d$ and

$W_R = Q_R$ or B_R , the cells based at points of $t + \Lambda$ in $z + W_R$ contain $z + W_{R-d}$ up to a null set and are contained in $z + W_{R+d}$. Thus

$$\text{vol}(W_{R-d}) \leq \det \Lambda \#((t + \Lambda) \cap (z + W_R)) \leq \text{vol}(W_{R+d}).$$

Divide by $\text{vol}(W_R)$ and sum over the distinct cosets. Every stated center-count intensity is therefore $|T|/\det \Lambda$, uniformly in z ; the corresponding covering density is $V|T|/\det \Lambda$. The finite constant d may depend on the lattice; no bound uniform over lattices is needed. Since periodic center sets are locally finite, restricting each unrestricted infimum to these competitors proves the reverse inequality. $\square$

Average overlap multiplicity. One customary definition uses the lower ball density of the overlap multiplicity [11, Section 1.1]. It agrees with the convention above. For either choice of W_R , put

$$A_W(X, R) = \sum_{x \in X} \text{vol}((K + x) \cap W_R).$$

Choose $s > 0$ such that K is contained in the radius- s ball of the norm defining W_R . Local finiteness gives, for $R > s$,

$$V\#(X \cap W_{R-s}) \leq A_W(X, R) \leq V\#(X \cap W_{R+s}).$$

Replacing R by $R+s$ or $R-s$ does not change the liminf or limsup of $\#(X \cap W_R)/\text{vol}(W_R)$: substitute the shifted radius and use $(R \pm s)^n/R^n \rightarrow 1$. The lower and upper limits of $A_W(X, R)/\text{vol}(W_R)$ therefore equal the corresponding center-count intensities multiplied by V . This counts each overlapping translate separately; the volume of the covered union is not the density.

Transfer of the lower bound. In particular, any inequality $\theta_{\text{per}}(K) > R$ established by the finite-coset argument implies $\theta_T(K) > R$ for the same body and the same constant. This uses equality of the infima and makes no claim that an individual covering can be repeated at exactly its original density. For every $\epsilon > 0$, a covering of finite density D does imply existence of a periodic covering of density less than $D + \epsilon$.

References

- [1] Boris Bukh, Jun Gao, Xizhi Liu, Oleg Pikhurko, and Shumin Sun. Covering large-dimensional Euclidean spaces by random translates of a given convex body. arXiv:2510.25685v2, 2025. Version 2, 17 August 2026; first posted 29 October 2025.
- [2] H. S. M. Coxeter, L. Few, and C. A. Rogers. Covering space with equal spheres. *Mathematika*, 6(2):147–157, 1959.
- [3] Ilya Dumer. Covering spheres with spheres. *Discrete & Computational Geometry*, 38(4):665–679, 2007. Corrected version: arXiv:math/0606002v2, 20 May 2018.
- [4] Gábor Fejes Tóth. A note on covering by convex bodies. *Canadian Mathematical Bulletin*, 52(3):361–365, 2009.
- [5] T. E. Harris. A lower bound for the critical probability in a certain percolation process. *Proceedings of the Cambridge Philosophical Society*, 56(1):13–20, 1960.

- [6] Svante Janson. Poisson approximation for large deviations. *Random Structures & Algorithms*, 1(2):221–229, 1990.
- [7] Svante Janson and Lutz Warnke. The lower tail: Poisson approximation revisited. *Random Structures & Algorithms*, 48(2):219–246, 2016.
- [8] Heng Li and Xizhi Liu. Nearly sharp bounds for lattice coverings by convex bodies. [arXiv:2607.28429v2](https://arxiv.org/abs/2607.28429v2), 2026. Version 2, 10 August 2026.
- [9] Márton Naszódi. Flavors of translative coverings. In *New Trends in Intuitive Geometry*, volume 27 of *Bolyai Society Mathematical Studies*, pages 335–358. Springer, 2018.
- [10] OpenAI. A single-lattice covering bound of order $n \log n$. OpenAI Math Release preprint [OAI:A-single-lattice-covering-bound-of-order-n-log-n-September-23-2026](#), 2026. Companion manuscript, Theorem 1.1.
- [11] Roman Prosanov. On a relation between packing and covering densities of convex bodies. *Discrete & Computational Geometry*, 65(4):1028–1037, 2021.
- [12] C. A. Rogers. A note on coverings. *Mathematika*, 4(1):1–6, 1957.